SongNet: Time-Resolved Music Genre Classification

UE24CS352A Machine Learning Mini-Project | Project Write-up
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https://github.com/nidhi832/SONGNET

1. Problem Statement and Motivation

Music genre classification supports playlist generation, recommendation and audio indexing. Classical pipelines describe a whole track with one vector of hand-crafted summary features and feed it to a shallow classifier, so how the sound evolves over time is lost. We re-implement SongNet [1], a convolutional-recurrent network (CRNN) that classifies 30-second clips into eight genres from log-Mel spectrograms and outputs a genre distribution at every time step. We compare two SongNet variants against five classical baselines on FMA-Small [2], and use the per-timestep outputs to visualise how genre confidence changes across a clip. These outputs provide a foundation for real-time genre tracking; streaming latency was not benchmarked in this work.

2. Dataset

FMA-Small [2] contains 8,000 balanced 30-second tracks (1,000 per genre): Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop and Rock. During preprocessing three MP3 files were corrupt and skipped, so **7,997 tracks** were used. Audio is resampled to 22.05 kHz and converted to log-Mel spectrograms (128 Mel bins, FFT size 2048, hop length 512), giving a 128 × 1292 input per track. A stratified 70/20/10 split gives **5,597 training, 1,600 validation and 800 test tracks** (100 per genre in test). Normalisation statistics are computed on the training split only, so no test information leaks into preprocessing.

3. Approach and Architecture

A spectrogram is not shift-invariant along frequency (a pitch change moves energy to different bins) but is along time. SongNet therefore treats the 128 Mel bins as input channels and convolves only along time. Both variants share one backbone and differ only in the classification head.

- **Shared backbone:** three temporal Conv1D blocks (128→128, 128→256, 256→256 channels, kernel size 5), each Conv1d → BatchNorm → ReLU → MaxPool(2) → Dropout(0.3). The 1292 input frames are reduced to a 256 × 161 feature map.
- **Variant A, Time-Distributed head:** a 1×1 Conv1d classifier applied independently at each of the 161 time steps. There is no recurrence.
- **Variant B, GRU / CRNN head:** a unidirectional GRU (hidden size 128) runs over the feature sequence, followed by a linear classifier at each step, so each prediction depends on earlier context. CRNN denotes a Convolutional Recurrent Neural Network.
- **Song-level prediction:** per-step softmax probabilities are averaged over time to give the song-level genre probabilities, which the loss and the test metrics use.
- **Baselines:** kNN (k=5), Logistic Regression, MLP (100, 50), Linear SVM and Random Forest (200 trees), trained on the hand-crafted features provided with FMA (features.csv) and standardised.

4. Implementation Overview

The code is a PyTorch pipeline. `src/preprocess.py` builds spectrograms, the split and normalisation statistics; `src/baselines.py` trains the classical models; `src/model.py` defines SongNet; `src/train.py` trains it; `src/evaluate.py` reports metrics and plots; `tests/smoke_test.py` is a quick synthetic end-to-end check. The Time-Distributed model was trained with `python -m src.train --data_dir data/processed --out_dir runs/songnet_full --epochs 20`. The GRU variant was trained with the same script (--head gru).

- **Optimisation:** Adam (learning rate 1e-3, weight decay 1e-4) with ReduceLROnPlateau (factor 0.5, patience 3); loss is the negative log-likelihood of the mean-pooled step probabilities.
- **Data:** batch size 32, dropout 0.3, random 640-frame (≈14.9 s) training crops as augmentation, CUDA GPU. Test evaluation uses the full-length spectrograms.
- **Model selection:** the checkpoint is saved whenever validation loss or validation accuracy improves. The Time-Distributed run reached a best validation accuracy of 55.75% and a lowest validation loss of 1.2900.
- **Demo:** the Gradio app (`python app.py`, default port 7860) accepts an uploaded or recorded clip and shows per-timestep genre probability curves, a temporal heatmap and song-level genre probabilities.

5. Results and Per-Genre Analysis

On the 800 test tracks, the GRU / CRNN model reached **59.75% accuracy and 0.5897 macro F1** (weighted F1 also 0.5897). The best classical baseline is Logistic Regression (56.12%, macro F1 0.5605), so the GRU / CRNN model is **3.63 percentage points** higher in accuracy and 0.0292 higher in macro F1. The Time-Distributed variant (56.25%) is about level with Logistic Regression. Random guessing over eight classes gives an expected 12.5%.

Table 1. Test results (800 tracks)

Model	Input	Test accuracy	Macro F1
SongNet, GRU / CRNN	**Log-Mel spectrogram**	**59.75%**	**0.5897**
SongNet, Time-Distributed head	Log-Mel spectrogram	56.25%	0.5561
Logistic Regression	FMA features	56.12%	0.5605
Random Forest	FMA features	55.38%	0.5467
MLP Classifier	FMA features	54.50%	0.5434
Linear SVM	FMA features	50.88%	0.5105
kNN (k=5)	FMA features	46.38%	0.4486

Table 2. GRU / CRNN per-genre test metrics (100 tracks per genre)

	Electronic	Experimental	Folk	Hip-Hop	Instrumental	International	Pop	Rock
Precision	0.67	0.42	0.57	0.79	0.51	0.67	0.42	0.68
Recall	0.64	0.47	0.79	0.78	0.41	0.68	0.28	0.73
F1	**0.66**	**0.44**	**0.66**	**0.78**	**0.46**	**0.68**	**0.34**	**0.71**

Hip-Hop is the strongest class (F1 0.78) and Pop the weakest (F1 0.34, recall 0.28). Experimental (0.44) and Instrumental (0.46) are also below 0.5. These genres are broad and varied, so they are the most likely to overlap with neighbouring classes; the confusion matrix produced by src/evaluate.py shows the exact pairs. Folk has high recall (0.79) but lower precision (0.57), meaning other genres are often labelled Folk.

6. Conclusion and Future Work

A CRNN with a GRU head reaches 59.75% test accuracy and 0.5897 macro F1 on FMA-Small, ahead of the best classical baseline (Logistic Regression, 56.12% and 0.5605) and of the Time-Distributed ablation (56.25% and 0.5561). The margin is modest and comes from a single run.

References

[1] C. Zhang, Y. Zhang and C. Chen, “SongNet: Real-time Music Classification,” Stanford University, CS229 course project report, 2018.
[2] M. Defferrard, K. Benzi, P. Vandergheynst and X. Bresson, “FMA: A Dataset for Music Analysis,” Proc. ISMIR, 2017.